



Separation of isopropyl ether and acetone using ionic liquids based on quantum chemistry calculation and liquid–liquid equilibrium



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ABSTRACT

Selecting green and efficient extractive solvents are the inevitable requirement of sustainable development. Ionic liquids as green solvents are widely used in the separation of organic substance. In this manuscript, ionic liquids are used as extractive solvents to separate isopropyl ether–acetone system. The selected ionic liquids are 1-ethyl-3-methylimidazole dinitrile amine [EMIM][DCA], n-butyl-3-methylimidazole trifluoromethane sulfonate [BMIM][OTF], 1-ethyl-3-methylimidazole trifluoromethane sulfonate [EMIM][OTF]. Firstly, the relation between H-bonding donor and H-bonding acceptor is description by COSMO-SAC model based on the extraction mechanism. After that the experiments of liquid–liquid equilibrium are conducted at $T = 298.15\text{ K}$ and $P = 101.325\text{ kPa}$, and the liquid–liquid equilibrium data are correlated by NRTL model. The results indicate that the model has a good correlation with the experimental results. Finally, the extraction mechanism is discussed at micro scale. The H-bonding interaction between ionic liquids and acetone, including total charge density, bond length, deformation charge density and interaction energy, is determined by quantum chemistry calculation. The quantum chemical calculation and experimental process provide theoretical guidance for the highly efficient separation of acetone – isopropyl ether binary system.

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1. Introduction

Isopropyl ether, also named as di-isopropyl ether (DIPE). It is a colorless liquid and is a good solvent for paraffin and resin. DIPE as a solvent is used in pharmaceutical, smokeless gunpowder, paint and paint cleaning, etc. In industry, the mixtures of DIPE and other solvents are often used in the dewaxing process of paraffin-based oil [1]. DIPE has high octane values and antifreeze properties which can be used as gasoline mixture [2]. Acetone is a vital raw material in chemical production. It is in great demand in chemical production [3]. It can not only be used as organic synthesis of raw materials [4,5], solvents, additives, such as the production of plexiglass, medicine, pesticides, epoxy resin, etc. [6,7], but also be used as extraction solvent in petroleum industry [8]. In the industrial process of catalytic dehydrogenation of isopropanol to acetone, a large number of DIPE by-products are obtained [9–11]. Separation of acetone/DIPE plays a key role in the production process, which affects the economic benefits of the production process. Efficient and economical separation of acetone and DIPE is the key to pro-

duce isopropyl alcohol. The traditional distillation process cannot separate acetone and DIPE owing to it forms a homogeneous azeotrope, but pressure swing distillation [12,13] extractive distillation [14,15] and azeotropic distillation [16,17] are all valid azeotropic separation ways. However, this process uses a good deal of energy. In contrast, liquid–liquid extraction [18–20] shows greater advantages, which use little energy. And ionic liquids (ILs) are odorless, nonflammable, extremely low vapor pressure [21], excellent heat resistance, easy recovery, and designed [21]. Meanwhile, it has selective solubility and an appropriate liquid phase scope [22,23]. What's more, they can be cyclic utilization by simple distillation [24]. This superiority makes ILs more and more widely used in liquid–liquid extraction. It is an ideal substitute for conventional extractive solvents and also is green and efficient solvents [25,26]. liquid–liquid extraction using ILs meets the demands of sustainable development and clean technology. So, extraction technology using ILs has been extensively accepted and recognized.

Because of the large number of ILs, researchers conduct preliminary screening before selecting suitable ILs. Firstly, the extractive solvents with better extraction effect of the separated DIPE + acetone binary azeotropic system is selected through pre-experiment.

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Then σ -profiles obtained by COSMO model. The mean can qualitatively determine the hydrogen bonding capacity of solvents and azeotropes. So as to further judge the extraction effect. Real solvents resemble conductor shielding models (COSMO-RS) provides a method for selecting suitable solvents [27]. The COSMO-RS model has high efficiency and accuracy in the calculation of dielectric shielding energy and dielectric shielding charge on van der Waals-like molecular surface. A new continuum model—the segmenting activity coefficient model (COSMO-SAC) proposed by Schürmann and Klamt on the basis of COSMO-RS is a new continuum model [28,29]. Sandler [30] proposes a method for predicting the heat of evaporation and vapor pressure using COSMO-SAC. An improved model is put forward to predict the thermodynamic properties of pure and mixed liquids, which greatly improves the accuracy of the prediction. With the development of quantum chemistry calculation, The COSMO model, which combines quantum chemistry calculation with statistical mechanics, has been proved to be a powerful tool for rapid screening of ILs [31–33]. Predicted the main steps of the COSMO model to obtain the σ -profiles. The σ -profiles can be used to confirm whether H-bonding can be formed between two molecules. The method can qualitatively determine the H-bonding capacity of the solvent and azeotrope, which provides prerequisite for the analysis of interaction. Meanwhile, the extraction capacity of different ILs is analyzed, and the interaction is verified by quantum chemistry calculation. Xu et al. [34] uses the COSMO-SAC model to study the selectivity of ILs during the extraction of coal tar fractions. It provides ideas for the separation of coal tar fractions, providing a thinking for the coal tar oil fraction separation. Meanwhile, the use of ILs in liquid–liquid equilibrium attracts much attention. Zhu et al. [35] use the liquid–liquid equilibrium method to study the azeotrope of acetone and cyclohexanone which use 1-butyl-3-methylimidazolium dicyanamide ([BMIM][N(CN)₂]), 1-butyl-3-methylimidazolium trifluoromethanesulfonate ([BMIM][NTf₂]) as extractive solvents. And extraction mechanism is discussed on the micro scale. Zhu et al. [36] separate acetone from n-heptane using 1-butyl-3-methylimidazolium trifluoromethanesulfonate ([BMIM][OTF]), 1-butyl-3-methylimidazolium bis (trifluoromethyl sulfonyl) imide ([BMIM][NTf₂]) and 1-butyl-3-methylimidazolium hexafluorophosphate ([BMIM][PF₆]). The extraction mechanism is analyzed. In the previous research, the separation of acetone and alkanes is mainly studied, focusing on the equilibrium data of liquid–liquid extraction. In order to further study the separation of acetone, acetone and ethers are separated based on previous work, and then deeply analyzed the extraction mechanism.

In this study, a method combining experiment and quantum chemistry calculation is proposed to separate acetone and DIPE system. The methods of pre-experiment and quantum chemistry calculating are used to choose the best extractive solvent. The σ -profiles are painted by the COSMO-SAC, and the relationship between H-bonding donor and acceptor is proved. Then the LLE data of {acetone DIPE and 1-n-butyl-3-methylimidazolium trifluoromethane sulfonate ([BMIM][OTF])}, acetone DIPE and 1-n-butyl-3-methylimidazolium trifluoromethane 1-ethyl-3-methylimidazolium trifluoromethanesulfonate ([EMIM][OTF]), acetone DIPE and 1-ethyl-3-methylimidazolium dinitrile amine ([EMIM][DCA]) is measured, and the extraction effect is evaluated by calculating the distribution coefficient (D) and selectivity coefficient (S). Meanwhile, NRTL method is used to finish the regression of experimental data, and the root mean square deviation (RMSD) of the experimental data and calculated data is calculated. Charge density, the bond length and interaction energy of DIPE and the ILs are calculated by the methods of quantum chemistry calculation and the liquid–liquid extraction mechanism is verified. The manuscript supplies a new theory instruction for selecting of extractive solvents for azeotrope mixtures.

2. Methods

2.1. COSMO-SAC and σ -profiles analysis

In the study, the extractive solvents with better extraction effect are selected in the azeotropic system of acetone and DIPE. The COSMO-SAC is used to verify the interaction between H-bonding receptor and H-bonding donor receptor [37,38], the model is used to determine the H-bonding between ILs in the acetone and DIPE system. The required compound information is shown in Table 1.

The steps for σ -profiles analysis are as follows: At first, download the initial chemical structure from the ChemicalBook or draw the chemical structure of each substance using GaussView 5.0. Then, On the basis of DFT theory and DMOL³ module, the chemical structure is optimized in Materials Studio 7.0. The GGA/VWN-BP function can be used. The basic value is set to DNP 3.5 and the actual space cut-off value is set to 5.5 Å [39]. After the molecular structure of geometric optimization of the molecular structure, the energy optimization is needed for each structural molecule, which ensure that the molecule is conceived as a minimum energy value. After optimization, a COSMO file is formed. Finally, Origin 2018 is used to draw σ -profiles. σ -profiles of DIPE, acetone and selected ILs as shown in the Fig. 1.

As shown in Fig. 1, Two vertical lines divide the σ -profiles into three areas and the σ -profile can reflect the surface charge distribution of molecules. When $\sigma < -0.0084$ e/Å², there represents H-bonding donator area, which means that the substance is capable of accepting the H-bonding. When $-0.0084 < \sigma < 0.0084$, there represents non-polar area. When σ greater than 0.0084 e/Å², there represents H-bonding acceptor area, which illustrates that the substance has the capacity to supply the H-bonding.

Based on the distribution region of different substance on the σ -profiles, the H-bonding receiving ability and H-bonding supplying ability of every molecule can be determined [40]. According to H-bonding donator and H-bonding acceptor relationship between ILs DIPE and acetone, DIPE can more easily bind to the extractive solvents by H-bonding. [EMIM][DCA] [EMIM][OTF] and [BMIM][OTF] have distinct areas in the H-bonding acceptor and donor regions, indicating the capacity to accept and provide hydrogen bonds. Acetone and DIPE have distinct areas in the H-bonding acceptor region, indicating the capacity to accept H-bonding. Owing to the strong H-bonding accept ability of acetone, acetone easily form H-bonding with selected ILs. It can be shown from Fig. 1, in the H-bonding donor area, [BMIM][OTF] has a largest area, the second is [EMIM][OTF], and the smallest is [EMIM][DCA]. It can be preliminarily determined that acetone is more likely to bind to [BMIM][OTF], followed by [EMIM][OTF], and finally [EMIM][DCA]. Therefore, selected ILs are suitable extractive solvents.

3. Experiments

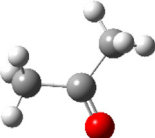
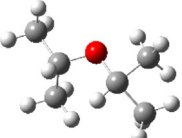
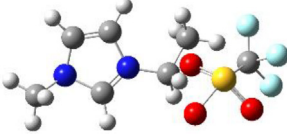
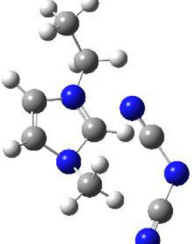
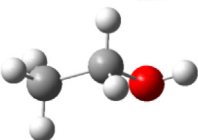
3.1. Chemicals

The information of acetone DIPE [EMIM][OTF] [EMIM][DCA] and [BMIM][OTF] is shown in Table 1. All solvents can be put to use immediately without further purification. Details of all compound structures, quality purity provided by the supplier and CAS number are shown in Table 1.

3.2. Instruments and procedure

{DIPE + acetone + [EMIM][OTF]}, {DIPE + acetone + [BMIM][OTF]}, {DIPE + acetone + [EMIM][DCA]} are measured by LLE experiment. The experiment is carried out at T = 298.15 K and P = 101.325 kPa. Instruments and detailed procedure for measuring can

Table 1Sources, mass purity stated by supplier information, of the compounds used in this study. (w_w is water contents in mass fractions).

Chemical	CAS number	Source	Structure	Molecular formula	Molar mass/ (g·mol ⁻¹)	Mass purity provided by supplier	$w_w/10^{-6}$	Analysis method
acetone	67-64-1	Sinopharm Chemical Reagent Co. Ltd		CH ₃ COCH ₃	58.08	0.995 ^a		GC
diisopropyl ether	108-20-3	Tianjin Damao Chemical Reagent Co., Ltd		C ₆ H ₁₄ O	102.17	0.990 ^a		GC
[BMIM] [OTf]	174899-66-2	Lanzhou Institute of Physical Chemistry		C ₉ H ₁₅ F ₃ N ₂ O ₃ S	288.29	0.980 ^a	< 500 ^a	
[EMIM] [OTf]	145022-44-2	Lanzhou Institute of Physical Chemistry		C ₇ H ₁₁ F ₃ N ₂ O ₃ S	260.23	0.980 ^a	< 500 ^a	
[EMIM] [DCA]	370865-89-7	Lanzhou Institute of Physical Chemistry		C ₈ H ₁₁ N ₅	177.21	0.980 ^a	< 500 ^a	
ethanol	64-17-5	Shanghai Aladdin Biochemical Technology Co. Ltd		C ₂ H ₆ O	46.07	0.998 ^a		GC

^aAnalysis by supplied.

be found in our previous work [41,42]. The experimental process is described as follows. ILs, DIPE and acetone are mixed into mixtures in different ratios and put into an equilibrium kettle. Stir the mixture vigorously with a stirring stick at approximately 1000 rpm. After stirring, the mixture was still for 54000 s approximately until it is absolutely separated. The cryostat (accurate to 0.05 K) maintains a constant temperature of 298.15 K during mixing and resting. Samples are taken for analysis after resting.

A Porapak Q column and a thermal conductivity detector (TCD) are used for gas chromatography (GC). Carrier gas is helium, the purity of which is greater than 99%. Ethanol as internal standard, ethanol, DIPE and acetone are measured through gas chromatography. The analyzed time of each sample should be no less than three times. Because ILs could not be detected by gas chromatography, the ILs composition is analyzed by method of mass difference. Sample is placed in a vacuum drying oven at 373.15 K and dried for 172800 s. Weighed the ILs in the sample after drying.

3.3. LLE correlation

The ternary phase diagram of DIPE + acetone + ILs is drawn by using the measured experimental data, the nonlinear least square

method (MATLAB R2012b programming) is used to determine the binary interaction parameters of NRTL model through data regression in Table 1. The experimental results and NRTL regression results are shown in Fig. 2–4, and the interaction parameters of binary system are shown in Table 3.

In data regression, the following equation is used to calculate liquid–liquid equilibrium is as follows:

$$\lambda_i^I x_i^I = \lambda_i^{II} x_i^{II} \quad (1)$$

where x_i^I represents the molar fraction of raffinate phase I, x_i^{II} represents the molar fraction of extraction phase II, λ_i^I represents the activity coefficients of raffinate phase I, λ_i^{II} represents the activity coefficients of extraction phase II.

In NRTL model, the interaction parameters of binary system are calculated by minimize the objective function value (OF) [43], and the equation is as:

$$OF = \sum_{k=1}^M \sum_{j=1}^2 \sum_{i=1}^3 (x_{ijk}^{exp} - x_{ijk}^{calc})^2 \quad (2)$$

where M represents the number of connecting lines, x^{exp} represents the experimental data of the mole fraction of the component, x^{cal} is

Table 2

Experimental data of DIPE (1) + acetone (2) + selected ILs (3) at P = 101.325 kPa and T = 298.15 K. (x_i = mole fraction, 1 2 3 represents DIPE, acetone and selected ILs, respectively). The data in Table 2 are the tie-line data in Figs. 2-4).

DIPE-rich Phase		ILs-rich phase				D	S
x ₁	x ₂	x ₃	x ₁	x ₂	x ₃		
DIPE (1) + acetone (2) + [BMIM][OTF] (3)							
0.9971	0.0000	0.0029	0.1325	0.0000	0.8675		
0.8347	0.1638	0.0015	0.0918	0.3754	0.5328	2.2918	20.8384
0.7330	0.2648	0.0022	0.1076	0.4301	0.4623	1.6242	11.0648
0.6276	0.3668	0.0056	0.1176	0.5805	0.3019	1.5826	8.4460
0.5409	0.4485	0.0106	0.1217	0.6073	0.2710	1.3541	6.0182
0.4252	0.5561	0.0187	0.1688	0.6861	0.1451	1.2338	3.1078
0.3331	0.6415	0.0254	0.2524	0.6937	0.0539	1.0814	1.4271
DIPE (1) + acetone (2) + [EMIM][OTF] (3)							
0.9985	0.0000	0.0015	0.0666	0.0000	0.9334		
0.8056	0.1937	0.0007	0.0435	0.3665	0.5900	1.8921	35.0408
0.6984	0.3000	0.0016	0.0351	0.4900	0.4749	1.6333	32.4991
0.6118	0.3838	0.0044	0.0495	0.5690	0.3815	1.4825	18.3236
0.5388	0.4588	0.0024	0.0563	0.6039	0.3398	1.3163	12.5968
0.4740	0.5071	0.0189	0.0769	0.6519	0.2712	1.2855	7.9239
0.4027	0.5833	0.0140	0.0685	0.7037	0.2278	1.2064	7.0923
DIPE (1) + acetone (2) + [EMIM][DCA] (3)							
0.9995	0.0000	0.0005	0.0650	0.0000	0.9350		
0.8276	0.1723	0.0001	0.0165	0.2804	0.7031	1.6274	81.6261
0.7097	0.2896	0.0007	0.0251	0.3973	0.5776	1.3719	38.7901
0.6181	0.3779	0.0040	0.0204	0.4727	0.5069	1.2509	37.8998
0.5409	0.4589	0.0002	0.0233	0.5195	0.4572	1.1321	26.2802
0.4748	0.5237	0.0015	0.0292	0.5704	0.4004	1.0892	17.7103
0.4205	0.5759	0.0036	0.0436	0.5970	0.3594	1.0366	9.9979

^a Standard uncertainties u are $u(T) = 0.05$ K, $u(p) = 1.5$ kPa, $u(x_i) = 0.0119$.

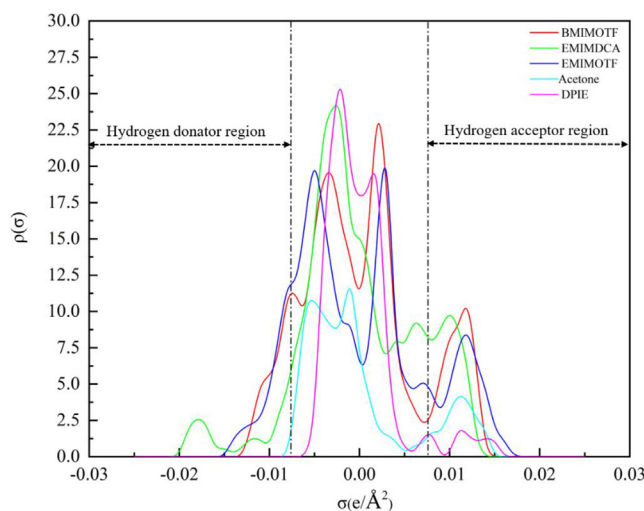


Fig. 1. σ -Profiles of acetone, DIPE, [EMIM][DCA], [EMIM][OTF] and [BMIM][OTF].

the calculated data of the mole fraction of the component, j represents the number of phases, i represents the composition, and k represents the connecting line. The value of the objective function is set to 10^{-6} during the regression process.

In order to check whether the binary interaction parameters obtained are suitable for the studied system. RMSD is used to check, and the equation is as follows [44]:

$$rmsd = \left(\sum_{k=1}^M \sum_{j=1}^2 \sum_{i=1}^3 \frac{(x_{ijk}^{exp} - x_{ijk}^{calc})^2}{6M} \right)^{1/2} \quad (3)$$

where M represents the number of connecting lines, x^{exp} represents the experimental data of the mole fraction of the component, x^{calc} is the calculated data of the mole fraction of the component, i represents the composition, j represents the number of phases, and k represents the connecting line. The value of the objective function is set to 10^{-6} during the regression process.

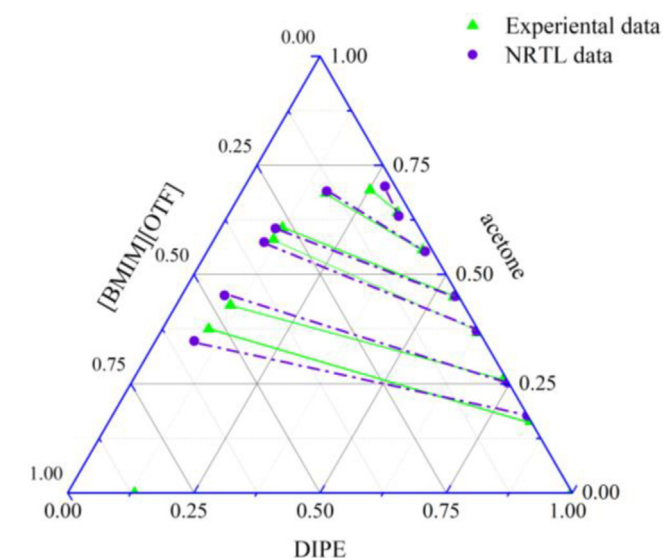


Fig. 2. Experimental and regression data of DIPE + acetone + [BMIM][OTF] at 101.325 kPa and 298.15 K. (▲—), experimental data; (●—), NRTL regression data.

resents the connecting line. The value of the objective function is set to 10^{-6} during the regression process.

The RMSD of NRTL in DIPE (1) + acetone (2) + [BMIM][OTF] (3), DIPE (1) + acetone (2) + [EMIM][OTF] (3) and DIPE (1) + acetone (2) + [EMIM][DCA] (3) is shown in Table 3.

3.4. Results and discussion

In extraction, because of the advantages of ILs in thermal stability, [EMIM][DCA], [BMIM][OTF] and [EMIM][OTF] are selected as extractive solvents. The viscosity, thermal stability and melting point of ILs are also important factors affecting the extraction pro-

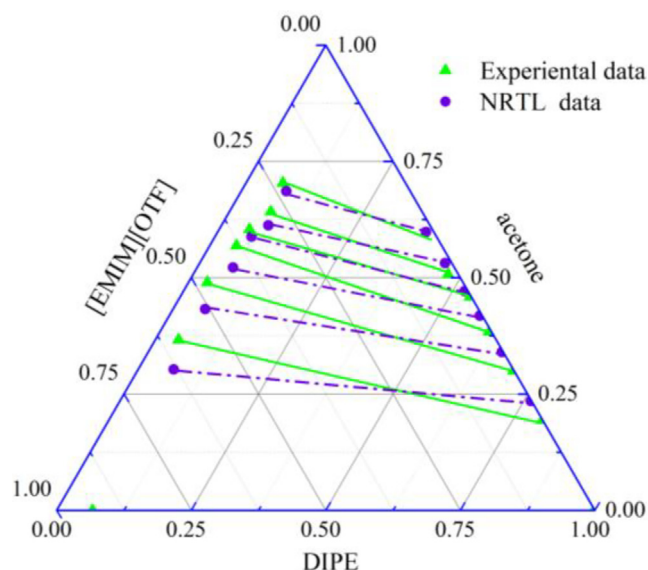


Fig. 3. Experimental and regression data of DIPE + acetone + [EMIM][OTF] at 101.325 kPa and 298.15 K. (▲—), experimental data; (●—), NRTL regression data.

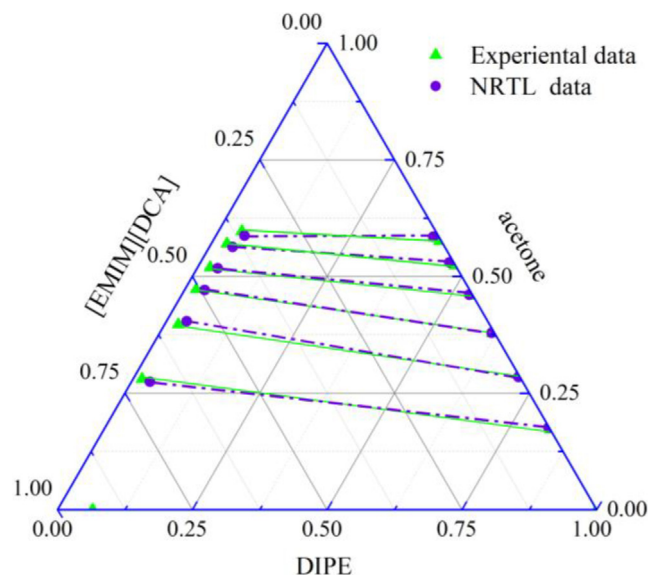


Fig. 4. Experimental and regression data of DIPE + acetone + [EMIM][DCA]

cess. The key physical properties of the selected ILs are shown in Table 4.

The LLE data of acetone + DIPE + [BMIM][OTF], acetone + DIPE + [EMIM][OTF] and acetone + DIPE + [EMIM][DCA] are determined at $P = 101.325$ kPa and $T = 298.15$ K, and the data are shown in Table 2. In Fig. 2–4, the tie lines of the ternary system are shown. The two layers of liquid at equilibrium are named as the conjugate phase. The line connecting the conjugate phases is named as the conjugate line. The immiscible region is the operating range of liquid–liquid extraction. From the ternary phase diagram, the selected ILs is suitable for DIPE and acetone. The immiscible region studied in this manuscript is large, which shows that it is reasonable to separate DIPE and acetone by liquid–liquid equilibrium.

S [52] and D [53] are important parameters in extraction process. D is an important parameter used to measure extraction capacity. S is the difference of extraction ability between two components. The smaller the D , the worse the extraction effect. The smaller the S , the more extractant used. The data of S and D are shown in Table 3. The changes of S and D with x_i^L are illustrate in Fig. 5–6.

As shown in Fig. 5, in the process of extraction and separation of acetone and DIPE, [BMIM][OTF] is better, followed by [EMIM][OTF] and [EMIM][DCA]. When D is low, it is necessary to comprehensively consider the influence of various factors on the extraction process. In the process of separating acetone and DIPE with ILs as extractant, D is small and similar, so it is necessary to comprehensively consider the influence of various factors. In the experiment, considering the effects of physical properties such as viscosity and D and S , the extraction efficiency of [EMIM][DCA] is the largest, followed by [BMIM][OTF] and [EMIM][OTF]. Therefore, [EMIM][DCA] is the best extraction solvent for the experimental part. In the systems of acetone + DIPE + [BMIM][OTF], acetone + DIPE + [EMIM][OTF] and acetone + DIPE + [EMIM][DCA], The results of calculating show that the values of S and D decrease with the increase of mole fraction in the system. At the same time, the value of S and D are all greater than 1, which indicates that it is viable to use the ILs of [BMIM][OTF], [EMIM][DCA] and [EMIM][OTF] to separate DIPE/acetone system. Where the formula of S and D is as follows:

$$S = \frac{(x_2^{IL-rich}/x_1^{IL-rich})}{(x_2^{DIPE-rich}/x_1^{DIPE-rich})} \quad (4)$$

$$D = \frac{x_2^{IL-rich}}{x_2^{DIPE-rich}} \quad (5)$$

where, subscript 1 is DIPE, subscript 2 is acetone, x_i is the mole fraction. The superscript ILs-rich is the extraction raffinate phase, the superscript DIPE-rich is the extraction phase.

Table 3

Data of the binary system are regressed by NRTL model at $P = 101.325$ kPa and $T = 298.15$ K.

$i-j$	t_{ij}	t_{ji}	α	RMSD
DIPE (1) + acetone (2) + [BMIM][OTF] (3)				
1–2	0.8073	1.6851	0.35	0.0090
1–3	7.1009	2.7925		
2–3	3.4403	–0.5131		
DIPE (1) + acetone (2) + [EMIM][OTF] (3)				
1–2	0.0650	2.1831	0.2	0.0158
1–3	7.3156	1.8210		
2–3	3.2373	–0.7767		
DIPE (1) + acetone (2) + [EMIM][DCA] (3)				
1–2	0.4518	2.2436	0.2	0.0077
1–3	7.5309	2.7652		
2–3	3.4869	–0.3698		

Table 4
Physical parameters of selected ILs.

	meting point (K)	decomposition temperature(K)	Density ^a (kg/m ³)	Viscosity ^a (Pa·s)
[BMIM][OTF]	289[45]	613[45]	1300[45]	0.084[46]
[EMIM][OTF]	262[47]	535[48]	1400[49]	0.043[49]
[EMIM][DCA]	267[50]	570[51]	1100[51]	0.015[51]

^a at 298 K and 101.3 kPa.

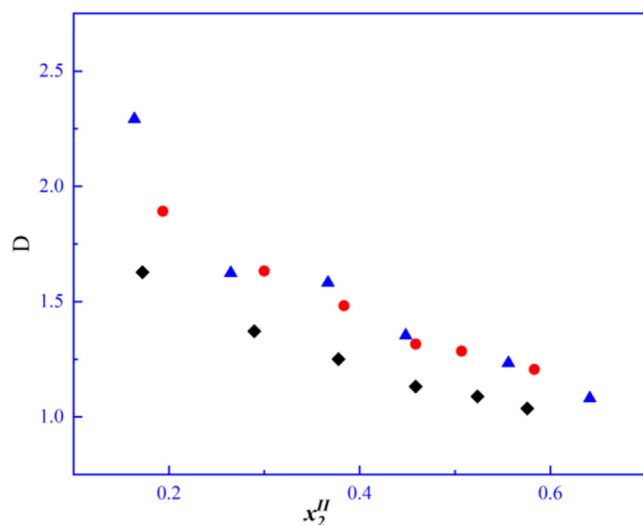


Fig. 5. The distribution coefficient of DIPE + acetone + selected ILs, (●), [EMIM][OTF] + DIPE + acetone; (▲), [BMIM][OTF] + DIPE + acetone; (◆), [EMIM][DCA] + DIPE + acetone.

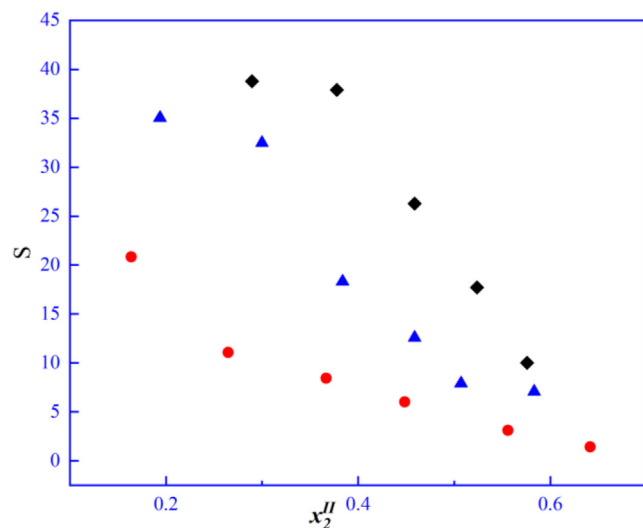


Fig. 6. The selectivity coefficient of DIPE + acetone + selected ILs, (●), DIPE + acetone + [BMIM][OTF]; (▲), DIPE + acetone + [EMIM][OTF]; (◆), DIPE + acetone + [EMIM][DCA].

As Table 3 and Figs. 2–4 shown, calculated data acetone + DIPE + [BMIM][OTF], acetone + DIPE + [EMIM][OTF] and acetone + DIPE + [EMIM][DCA] systems are consistent with the experimental data. The RMSD value obtained by calculation also shows that the deviation of the calculated and the experimental value is within 2%, which indicates that the interaction parameters of binary in NRTL model obtained by data regression can accurately describe the phase behavior of the study system [54], and also proves that NRTL is appropriate for thermodynamic simulation of DIPE + acetone + selected ILs.

4. Quantum chemical calculation

4.1. Interaction of acetone and ILs

In the manuscript, the extraction mechanism is illustrated at micro level. In order to study the interaction of DIPE-acetone and selected ILs, at first, based on optimizing the chemical structure and chemical energy, σ -profiles are obtained to analyze the interaction between the extracted substance and ILs. Next, the interaction between ILs and acetone system is calculated by using DMOL³ module in the Material Studio 7.0 software. Parameters of process are set to DNP and GGA/VWN-BP function is set to 3.5. All energy of molecular structures must be the minimum, so every molecule must be optimized. After the optimization of each molecule, different ILs need to be combined with acetone respectively and further optimized. The optimal configurations which are the complexes between selected ILs and acetone are obtained, and the complexes bond lengths are obtained. As Fig. 7 shown, the color of hydrogen atom is in the white, the color of oxygen atom is in the red, the color of nitrogen atom is in the dark blue, the color of fluorine atom is in the light blue, and the color of sulfur atom is in the yellow. The van der Waals radii between oxygen and hydrogen are 1.520 and 1.200 severally [54]. The van der Waals radius sum between oxygen and hydrogen is 2.720. When [BMIM][OTF] is used as extractive solvent, The length of the hydrogen and the oxygen atom in acetone is 1.961. When [EMIM][OTF] is used as extractive solvent, The distance between hydrogen atom and oxygen atom in acetone is 2.039. When [EMIM][DCA] is used as extractive solvent, The length of the hydrogen and the oxygen atom in acetone is 2.085. The van der Waals radius is greater than the distance. When the van der Waals radius is greater than the distance between atoms, there is an intermolecular interaction between selected ILs and acetone, which indicates that there is hydrogen bonding between selected ILs and acetone [55].

4.2. Charge density distribution

The charge density figure can be used for judging the size of the interaction force. The higher the isovalue between ILs and acetone, the stronger the interaction between acetone and extractive solvent. Fig. 8–10 shows that when the isovalue is 0.16, the charge density of acetone and [BMIM][OTF] still exists, which the strong interaction of ILs and DIPE is more distinct. when the isovalue is 0.15, the charge density of acetone and [EMIM][OTF] still exists. When the isovalue is 0.15, the charge density between [EMIM][DCA] and acetone exists, which the strong interaction of [BMIM][OTF] and DIPE is more distinct. However, the interaction of [EMIM][DCA], [EMIM][OTF] and acetone is not obvious, respectively.

4.3. Deformed charge density

The deformation charge density can be used to judge whether system is H-bonding of acetone and the selected ILs. As shown in Fig. 11, The red regions around O tend to accept charge, However, the blue regions around H atoms tend to lose charge. The junction of the blue and red regions shows formation of H-bonding. The

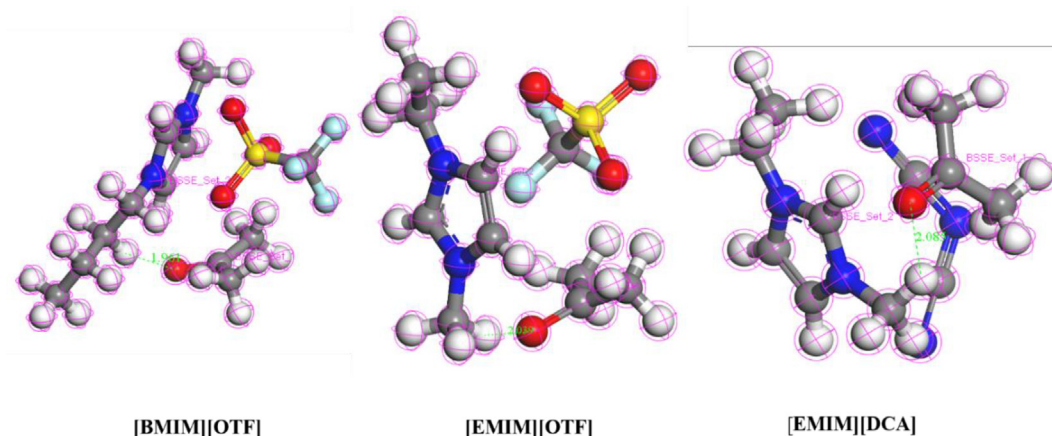


Fig. 7. Hydrogen bond lengths for acetone and selected ILs.

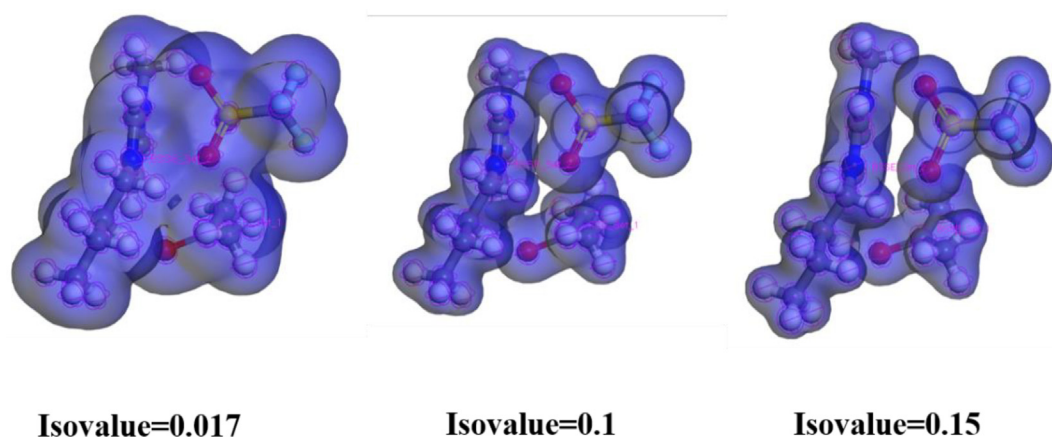


Fig. 8. Charge density distribution of acetone and [BMIM][OTF]

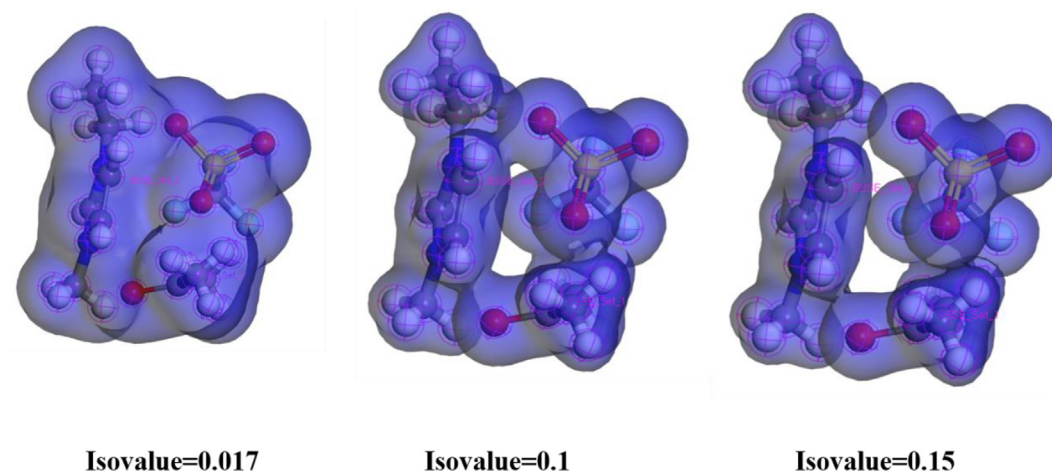


Fig. 9. Charge density distribution of acetone and [EMIM][OTF]

results show that it is likely to form H-bonding between acetone and the selected ILs.

4.4. Interaction energy

The interaction between molecules is illustrated by using the interaction energy. The interaction energy is shown in Table 4. The conformational energy formula is shown as follows [54]:

$$\Delta E = E_{AB} - E_A - E_B + E_{BSSE} \quad (6)$$

$$E_{BSSE} = E_A - E_{(A,AB)} + E_B - E_{(B,AB)} \quad (7)$$

where E_{AB} represents the interaction energy between the molecules, E_A and E_B represents the energy of DIPE and selected ILs, $E_{(B,AB)}$ is the energy of ILs in the binary system. ΔE is interaction energy of system, E_{BSSE} is correction energy.

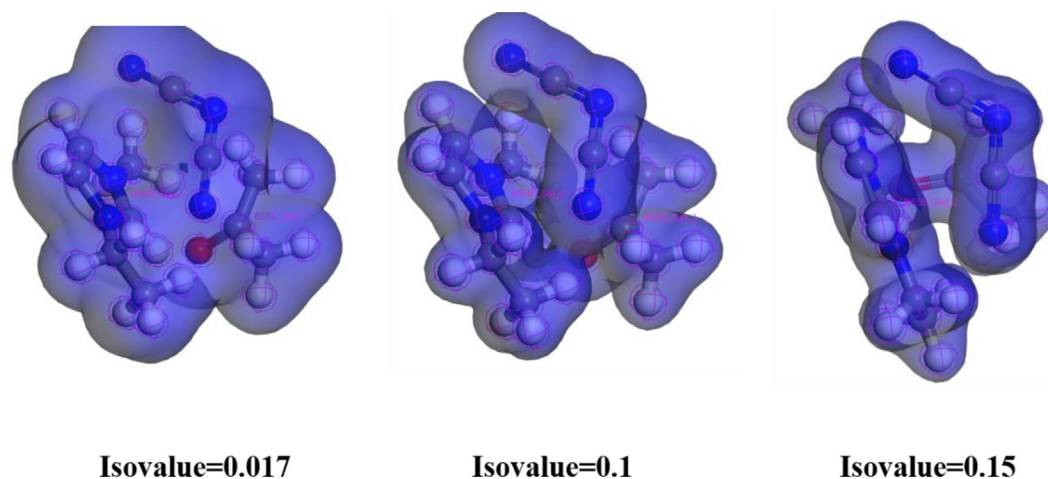


Fig. 10. Charge density distribution of acetone and [EMIM][DCA]

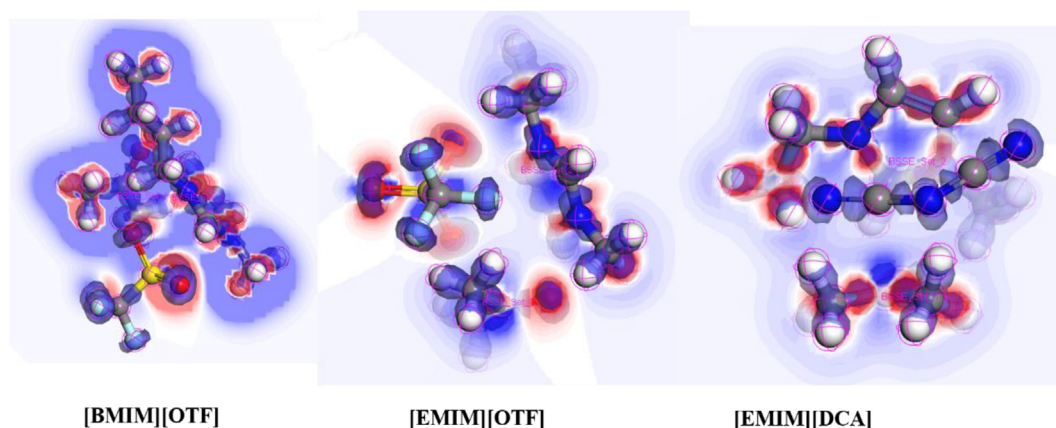


Fig. 11. Deformation charge density between acetone and selected ILs.

The energy of acetone with [EMIM][OTF], [BMIM][OTF] and [EMIM][DCA] are -54.6157 , -49.6924 and -39.9351 $\text{J}\cdot\text{mol}^{-1}\times 10^3$, respectively. The results indicate that acetone has the strongest interaction with [BMIM][OTF], followed by [EMIM][OTF], finally, [EMIM][DCA]. Detailed calculation data are shown in Table 5. The molecular extraction mechanism of the calculated results is the same as the experimental results. Thereby verifying the experimental results.

5. Conclusion

In this manuscript, the extraction of DIPE from acetone is analyzed by liquid–liquid experiments and quantum chemistry. The relation between H-bonding donor and H-bonding acceptor is illustrated by σ -profiles. The experiments DIPE + acetone + ILs ternary system at $T = 298.15$ K and $P = 101.325$ kPa, and the separation of acetone DIPE with [EMIM][OTF], [EMIM][DCA] and [BMIM][OTF] has good separation effect. S and D are vital parameters to estimate the extraction effect. The Table 2 shows the experimental results. And Fig. 5–6 are showed detailed data. The accuracy of the experiment is verified by NRTL. The extraction capacity of [EMIM][DCA] < [BMIM][OTF] < [EMIM][OTF] is obtained by the experimental data. In order to explain the mechanism of extraction in more detail, the simulation of extraction effect is added from the molecular point of view. The H-bonding between acetone and the selected ILs is verified based on calculating of

Table 5

Interaction energies of DIPE, acetone, selected ILs and DIPE and selected ILs (E is energy of substance).

Substance	E ($\text{J}\cdot\text{mol}^{-1}\times 10^8$)	E_{BSSE} ($\text{J}\cdot\text{mol}^{-1}\times 10^3$)	ΔE ($\text{J}\cdot\text{mol}^{-1}\times 10^3$)
DIPE	−8.2019		
acetone	−5.0734		
[BMIM][OTF]	−36.3736		
[EMIM][OTF]	−34.3086		
[EMIM][DCA]	−13.3677		
DIPE + [BMIM][OTF]	−41.4374	−3.8047	−54.6157
DIPE + [EMIM][OTF]	−39.3720	−4.4191	−49.6924
DIPE + [EMIM][DCA]	−20.4315	−7.1420	−39.9351

the DMOL³ module. The interaction and H-bonding of acetone and selected ILs is calculated. Bond length reflects the size of the interactions between molecules. The longer the bond distance, the worse the interaction formed, and the extraction effect of ionic liquid is worse. the worse the extraction efficiency of ILs. The smaller the absolute value of the interaction energy, the smaller the interaction, the worse the extraction capacity of ILs on acetone. Above all, [BMIM][OTF] has the best extraction capacity. The size of the interaction energy between acetone and selected ILs is [EMIM][DCA] < [EMIM][OTF] < [BMIM][OTF]. The manuscript mainly studies the experiment and mechanism analysis of acetone and DIPE. Therefore, the process simulation is given in the Supporting information. However, in the extraction of acetone and DIPE,

the bond length difference is small. So, it is the most advantageous way to determine the extraction effect through experiments. Therefore, the study provides a green and efficient extractive solvent for the separation of acetone and DIPE. The results show that the extraction and separation of azeotropic system with ionic liquids is a promising extraction technology. Therefore, ILs can be an option for the separation of azeotropes.

CRediT authorship contribution statement

Yanli Zhang: Conceptualization, Methodology, Investigation, Writing – original draft. **Huiyuan Li:** Validation, Data curation, Visualization, Software, Writing – review & editing. **Haixia Li:** Validation, Formal analysis. **Rongli Shan:** Data curation, Visualization, Software. **Zhaoyou Zhu:** Validation, Formal analysis. **Yinglong Wang:** Writing – review & editing. **Jun Gao:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jct.2021.106715>.

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